

Course Code- 151608 Distributed Computing**3 0 0 3****Unit-1.0: Introduction to Distributed Computing**

Definition, Characteristics of distributed systems, Types of distributed architectures (client-server, peer-to-peer, multi-tier), Distributed system models, Advantages and challenges, Design issues.

Unit-2.0: Inter-Process Communication & Synchronization

Message passing mechanisms, Remote Procedure Call (RPC), Remote Method Invocation (RMI), Synchronization in distributed systems, Logical clocks (Lamport & Vector clocks), Mutual exclusion algorithms (Ricart-Agrawala, Maekawa's algorithm).

Unit-3.0: Distributed Algorithms and Fault Tolerance

Leader election algorithms (Bully, Ring), Distributed consensus (Paxos, RAFT), Load balancing strategies, Distributed deadlock detection, Fault tolerance models, Checkpointing and recovery techniques.

Unit-4.0: Distributed File Systems and Databases

Distributed file systems (NFS, AFS, Google File System), Distributed databases (CAP theorem, NoSQL, Replication & Consistency models), Two-phase & three-phase commit protocols, Cloud-based storage architectures.

Unit-5.0: Distributed Computing Frameworks & Middleware

Hadoop ecosystem (HDFS, MapReduce, YARN), Apache Spark, Distributed middleware (CORBA, JINI, Web services), Edge and Fog Computing, Serverless computing.

Unit-6.0: Emerging Trends and Case Studies

Microservices and Kubernetes, Blockchain in distributed environments, AI and ML in distributed systems, Internet of Things (IoT) and distributed processing, Case studies on real-world distributed applications (Google Spanner, Amazon DynamoDB).

Text/Reference:

1. Andrew S. Tanenbaum & Maarten Van Steen, Distributed Systems: Principles and Paradigms, Pearson.
2. George Coulouris, Jean Dollimore, Tim Kindberg, Distributed Systems: Concepts and Design, Pearson.
3. Mukesh Singhal & Niranjana G. Shivaratri, Advanced Concepts in Operating Systems: Distributed, Database, and Multiprocessor Operating Systems, McGraw Hill.
4. Ajay D. Kshemkalyani & Mukesh Singhal, Distributed Computing: Principles, Algorithms, and Systems, Cambridge University Press.
5. Hagit Attiya & Jennifer Welch, Distributed Computing: Fundamentals, Simulations, and Advanced Topics, Wiley.
6. Tom White, Hadoop: The Definitive Guide, O'Reilly.
7. Martin Kleppmann, Designing Data-Intensive Applications, O'Reilly.